

EXPERIMENT - 6

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6.1.1] Incremented Date

A) ALGORITHM

1. Start.
2. Read day.
3. Read month.
4. Read year.
5. If $\text{year} \leq 0$, print "Invalid Date" and Stop.
6. If $\text{month} < 1$ or $\text{month} > 12$, print "Invalid Date" and Stop.
7. If month is 1, 3, 5, 7, 8, 10, 12 then $\text{max_days} = 31$.
8. Else if month is 4, 6, 9, 11 then $\text{max_days} = 30$.
9. Else if month is 2:
 If $(\text{year} \% 400 == 0)$ OR $(\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 \neq 0)$
 $\text{max_days} = 29$
 Else
 $\text{max_days} = 28$
10. If $\text{day} < 1$ or $\text{day} > \text{max_days}$, print "Invalid Date" and Stop.
11. If $\text{day} < \text{max_days}$ then $\text{day} = \text{day} + 1$.
12. Else
 $\text{day} = 1$
 If $\text{month} == 12$
 $\text{month} = 1$
 $\text{year} = \text{year} + 1$
 Else
 $\text{month} = \text{month} + 1$
13. Print day-month-year in format DD-MM-YYYY.
14. Stop.

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B) PYTHON CODE

```
day = int(input())

month = int(input())

year = int(input())


def is_leap(year):

    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):

        return True

    return False


def days_in_month(month, year):

    if month in [1, 3, 5, 7, 8, 10, 12]:

        return 31

    elif month in [4, 6, 9, 11]:

        return 30

    elif month == 2:

        return 29 if is_leap(year) else 28

    else:

        return 0


if year <= 0 or month < 1 or month > 12:

    print("Invalid Date")

elif day < 1 or day > days_in_month(month, year):

    print("Invalid Date")

else:

    day += 1

    if day > days_in_month(month, year):

        day = 1

        month += 1

        if month > 12:

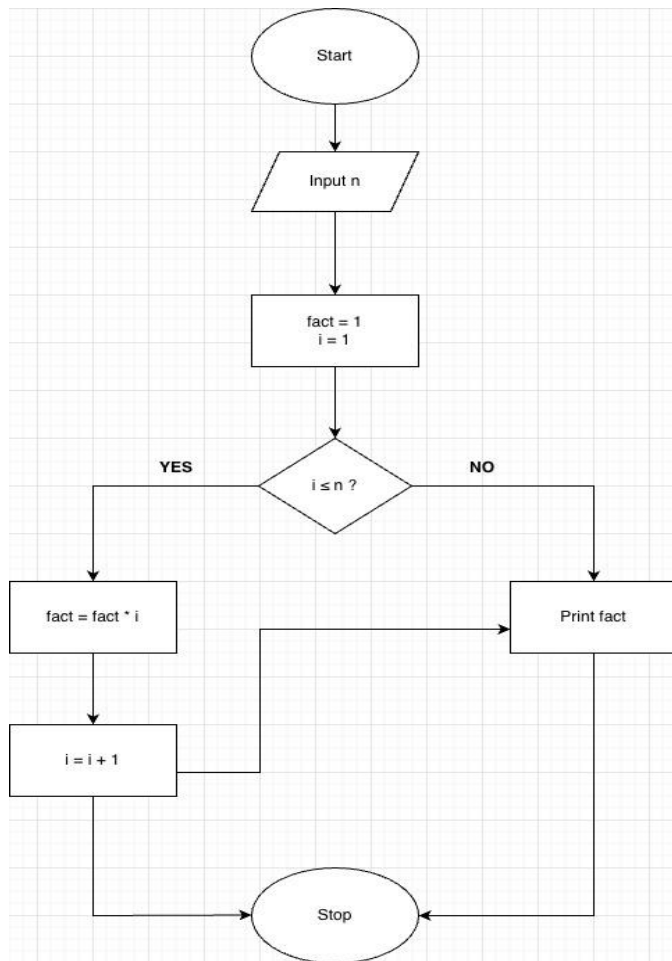
            month = 1

            year += 1

    print(f"{day:02d}-{month:02d}-{year}")
```

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C) FLOWCHART



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D) EXCECUTION

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6.1.1. Incremented Date 01:39

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. Valid Month: 1 to 12

2. Valid Day: Depends on the month and year

- January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
- April (4), June (6), September (9), November (11): 1 to 30 days
- February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year

3. Valid Year: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:
<DD>-<MM>-<YYYY>
- If the date is invalid, print: "Invalid Date".

Sample Test Cases +

nextDate.py

1 # Read input
2 day = int(input())
3 month = int(input())
4 year = int(input())
5
6 # Function to check leap year
7 def is_leap(year):
8 if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
9 return True
10 return False
11
12 # Function to get days in a month
13 def days_in_month(month, year):
14 if month in [1, 3, 5, 7, 8, 10, 12]:
15 return 31
16 elif month in [4, 6, 9, 11]:
17 return 30
18 elif month == 2:
19 return 29 if is_leap(year) else 28
20
21 # Main logic
22 if __name__ == '__main__':
23 day = int(input())
24 month = int(input())
25 year = int(input())
26 if not (1 <= month <= 12 and 1 <= day <= days_in_month(month, year) and year > 0):
27 print("Invalid Date")
28 else:
29 day += 1
30 if day > days_in_month(month, year):
31 day = 1
32 month += 1
33 if month > 12:
34 month = 1
35 year += 1
36 print(f"{day}-{month}-{year}")

Average time 0.004 s
Maximum time 0.007 s
4.00 ms 7.00 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1 5 ms

Expected output
15
3
2024
16-03-2024

Actual output
15
3
2024
16-03-2024

Test case 2 4 ms

Terminal Test cases

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